STAT 208: Statistical Learning Theory **Winter 2018**

Lecture 12 — 15 Feb. 2018

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This is the danger environment.
Warning: These scribe notes have only been mildly proofread.

Convex Surrogate Loss functions

Recall that $R(f) = \mathbb{E}\ \mathbb{I}\{Y \neq f(X)\}$ is the risk for the 0-1 loss, where $\mathcal{Y} = \{0, 1\}$, $f : \mathcal{X} \to \mathcal{Y}' = \mathbb{R}$ and suppose that we are interested in classification rules of the form $\widetilde{f} = \text{sign}(f(x))$.

Definition: A loss function $L : \mathcal{Y} \times \mathcal{Y}' \to [0, \infty)$ is said to be *Margin based* if there exists a ϕ such that

$$L(y, t) = \phi(yt)$$

Hence,

$$R(f) = \mathbb{P}(Y \neq \text{sign}(f)) = \mathbb{P}(Yf(X) \leq 0) = \mathbb{E}[\mathbb{I}\{Yf(X) \leq 0\}] = \mathbb{E}[L(Y, f(X))],$$

where $L(y, t)$ is the margin based loss $\mathbb{I}\{yt \leq 0\} := \phi_0(yt)$.
Other examples:
1. Hinge loss: $\phi(t) = (1 - t)_+ = \max(0, 1 - t)$
2. Squared hinge loss: $\phi(t) = (1 - t)_+^2$
3. Logistic loss $\phi(t) = \log(1 + e^{-t})$

We want to replace ϕ_0 with something smooth and hopefully convex, say ϕ. This is a surrogate loss function which has nice properties, perhaps tractable optimization etc. Moreover we want to relate $R_\phi(f)$ to $R_{\phi_0}(f)$ and answer whether minimum surrogate risk can approximate the minimum risk.

Define $\eta(X) = \mathbb{P}(Y = 1 \mid X = x)$, i.e., the optimal Bayes rule for the 0-1 loss $\mathbb{I}\{Y \neq \text{sign}(\eta(X))\}$. Define the Posterior loss for η as

$$C_\eta(t) := \mathbb{E}_\eta[L(Y, t)] \qquad \text{and} \qquad C_{\eta(x)}(t) := \mathbb{E}_\eta[L(Y, t) \mid X = x].$$

Observe that $R(f) = \mathbb{E}C_{\eta(X)}(f(X))$

Classification with Margin based loss

Let $L(y,t) = \phi(ty)$ whereby $C_\eta(t) = \eta\phi(t) + (1-\eta)\phi(-t)$, and let $t^*(\eta)$ be the minimizer of the functional $t \longmapsto C_\eta(t)$ and let $f^*_\phi := t^*_\phi(\eta)$. Furthermore define

$$C^*_\eta := \inf_{t\in\mathbb{R}} C_\eta(t),$$

whereby the Bayes risk is $R^* = R(f^*) = \mathbb{E}[C^*_\eta]$.

We want to bound $R_{\phi_0}(f) - R^*_{\phi_0} := R(f) - R^*$ in terms of $R_\phi(f) - R^*_\phi := \widetilde{R}(f) - \widetilde{R}^*$

Now, if ϕ is continuous (and differentiable) then $C_\eta(\cdot)$ is continuous (and differentiable). Moreover,

$$C_\eta(0) = (2\eta - 1)\phi'(0).$$

For simplicity we drop the dependence on η wherever it is unambiguous from the context.

Let $\mathcal{M}(\epsilon) = \mathcal{M}_\eta(\epsilon) = \{t \in \mathbb{R} \mid C_\eta(t) - C^*_\eta \le \epsilon\}$ be the set of ϵ approximate minimizers of C. And consider the calibration function

$$\delta(\epsilon) := \inf_{t\notin\mathcal{M}_\eta(\epsilon)} \widetilde{C}(t) - \widetilde{C}^*.$$

Indeed we wish this quantity were large since that would meet ϕ is a better surrogate for ϕ_0. We also note that by standard convention in optimization, $\delta(\epsilon) = +\infty$ if $C^* = \infty$ or if $\mathcal{M}(\epsilon) = \mathbb{R}$.

Note that $C(t) - C^* \ge t \implies \widetilde{C}(t) - \widetilde{C}^* \ge \delta(\epsilon)$ for all $t \in \mathbb{R}$ and $\epsilon > 0$.

For any fixed t taking supremum over $\epsilon \le C(t) - C^*$, we get

$$\widetilde{C}(t) - \widetilde{C}^* \ge \sup_{\epsilon:\epsilon\le C(t)-C^*} \delta(\epsilon) = \delta(C(t) - C^*),$$

where the last equality follows since $\delta(\cdot)$ is an increasing function.

Now, define $\mathcal{M}(0+) := \cap_{\epsilon>0}\mathcal{M}(\epsilon) = \operatorname*{argmin}_t C(t)$

Lemma 12.1. *If $\widetilde{C}(\cdot)$ is convex and $\mathcal{M}(\epsilon) = (t^-_\epsilon, t^+_\epsilon)$ is some interval. Moreover, if we have $\widetilde{\mathcal{M}}(0+) \subseteq \mathcal{M}(0+)$, then*

$$\delta(\epsilon) = \min\{\widetilde{C}(t^-_\epsilon), \widetilde{C}(t^+_\epsilon)\} - \widetilde{C}^*.$$

Definition: $\widetilde{L}$ is calibrated for L with respect to a set of measures $\mathbb{Q}$ if $\delta_\eta(\epsilon) > 0$ for all $\eta \in \mathbb{Q}$ and $\epsilon > 0$.

Definition: $\widetilde{L}$ is uniformly calibrated for L with respect to $\mathbb{Q}$ if $\delta_{\mathbb{Q}}(\epsilon) = \inf_{\eta\in\mathbb{Q}} \delta_\eta(\epsilon) > 0$ for all $\epsilon > 0$.

Let $B_f := \|X \longmapsto C_{\eta(X)}(f(X))\|_\infty$. Q[1]. We work with the distributions of type $\mathbb{Q}$, i.e., $\mathbb{P}(Y \in \cdot \mid X = x) \in \mathbb{Q}$.

Theorem 12.2. *Assume $\widetilde{L}$ is uniformly calibrated for L with calibration function $\delta_{\mathbb{Q}}(\cdot)$ and let $\delta^{**}_{B_f} : [0, B_f] \to [0, \infty)$ be the Fenchel-Legendre Biconjugate of δ restricted to $[0, B_f]$. Assume that R^* and $\widetilde{R}^*$ are finite. Then,*

$$\delta^{**}_{B_f}(R(f) - R^*) \leq \widetilde{R}(f) - \widetilde{R}^*.$$

Proof: Assume $R(f) < \infty$, and observe the following sequence of inequalities.

$$\begin{aligned}
\delta(C_{\eta(X)}(t) - C^*_{\eta(X)}) &\leq \widetilde{C}_{\eta(X)}(t) - \widetilde{C}^*_{\eta(X)}, \qquad \text{for all} \quad X \\
\delta^{**}(C_{\eta(X)}(f(X)) - C^*_{\eta(X)}) &\overset{\text{(a)}}{\leq} \widetilde{C}_{\eta(X)}(f(X)) - \widetilde{C}^*_{\eta(X)} \\
\mathbb{E}[\delta^{**}(C_{\eta(X)}(f(X)) - C^*_{\eta(X)})] &\leq \mathbb{E}[\widetilde{C}_{\eta(X)}(f(X)) - \widetilde{C}^*_{\eta(X)}] \\
\delta^{**}(R(f) - R^*) &\leq \widetilde{R}(f) - \widetilde{R}^*,
\end{aligned}$$

where (a) uses the substitution $t = f(X)$ and $\delta^{**} \leq \delta$, and the last step uses the Jensens inequality for the convex function δ^{**}. $\square$

[1]Does $\mathbb{E}[B_f] = R(f)$